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A. PURPOSE: To provides guidelines for monitoring oral or intravenous sedation administered to patients undergoing invasive, manipulative, or diagnostic procedures or patients admitted with mechanical ventilations in Cardiac Ward etc.

B. SCOPE: Hospital wide

C. DEFINITIONS OF SEDATION

1. Minimal sedation (anxiolysis) :

   A drug-induced state during which patients respond normally to verbal commands. Although cognitive function and coordination may be impaired, ventilatory and cardiovascular functions are unaffected.

2. Moderate sedation/analgesia ("conscious sedation") :

   A drug-induced depression of consciousness during which patients respond purposefully to verbal commands, either alone or accompanied by tactile stimulation. No interventions are required to maintain a patent airway, and spontaneous ventilation is adequate. Cardiovascular function is usually maintained.

3. Deep sedation/analgesia :

   A drug-induced depression of consciousness during which patients cannot be easily aroused, but respond purposefully following repeated or painful stimulation. The ability to independently maintain ventilatory function may be impaired. Patients may require assistance in maintaining a patent airway and spontaneous ventilation may be inadequate.

   Cardiovascular function is usually maintained.

D. GOALS

1. To guard the patient's safety and welfare

2. To minimize physical discomfort, pain, or anxiety

3. To minimize negative psychological responses to treatment by providing sedation and analgesia, and to maximize the potential for amnesia

4. To control behavior.

5. in some children and uncooperative adults, to expedite the conduct of procedures which are not particularly uncomfortable but which require that the patient not move.

6. to return the patient to a state in which safe discharge, as determined by recognized criteria, is possible for outpatients or to return inpatients to pre-sedation status

E. PROTOCOL:

LOCATION

Moderate or deep sedation is practiced in various departments and locations within the organization.

- PROCEDURE

  All moderate or deep sedation will be ordered and supervised by a qualified physician/dentist. He/she will assume the responsibility of the patient until patient meets discharge criteria and/or returns to pre-sedation status.

- The licensed professional nurse (RN) responsible for managing the care of patients receiving moderate or deep sedation will complete and maintain competency in the skill.

- The chairman or medical director of each department administering moderate or deep sedation will be responsible for ensuring that departmental policies and procedures are applicable and consistent with this hospital policy.

- Emergency equipment must be immediately available to every location where moderate or deep sedation is administered or recovery occurs, and includes at least the following: defibrillator, suction device, oxygen, airway management tools, emergency drugs, intubation equipment, and EKG monitor.

- All patients must have an intravenous access secured prior to administration of moderate or deep sedation. Children under age 8 may receive only chloral hydrate up to a dose of 100 mg/kg or only oral midazolam up to a dose of 0.5 mg/kg without an intravenous access.

iii. Qualification of staff administering sedation in the hospital :

- The person **attending and monitoring** sedation is **differen**t from the person performing the procedure.

- Sedation of patient will be done **only** by qualified and trained medical and nursing staff.

- Doctors and nurses supervising, administering, and monitoring moderate or deep sedation are required to have proficiency in the delivery of sedation. These individuals are required to:

1. be familiar with proper dosages, administration, adverse reactions, and interventions for adverse reactions and overdoses

2. Know how to recognize an airway obstruction and possess knowledge of basic life support.

4. Assess total patient care requirements or parameters, including but not limited to respiratory rate, oxygen saturation, blood pressure, cardiac rate, and level of consciousness.
5. Following **Equipmen**t and **manpowe**r are available to rescue patients from a deeper level of sedation than that intended.

- Ventilator
- Syringe Pumps
- Multipara Monitors
- Defibrillator
- Central Oxygen
- Suction Machine

- ECG machine
- Laryngoscope
- Ambubag

Team of Anesthetists and Physician trained in critical care medicine.

F. PRE PROCEDURE ASSESSMENT:

All patients requiring sedation will have a pre procedure evaluation and assessment including, but not limited to:

1. history and physical performed by a physician/dentist
2. current medications, including any allergies
3. prior history of adverse reaction to sedation or anesthesia
4. NPO status
5. proper consent forms signed
6. vital signs: heart rate, blood pressure, respiratory rate, oxygen saturation, pain score, level of consciousness, temperature when applicable

The physician will decide that the patient is a suitable candidate for planned sedation based on assessment.

G. MONITORING DURING PROCEDURE

Monitoring of the patient is to be performed throughout the procedure and will include documentation of vital signs: heart rate

blood pressure
respiratory rate
oxygen saturation
level of consciousness
end-tidal CO2

1. prior to initiation of sedation.

2. reviewed every 5 minutes (minimum).

3. recorded every 5 minutes (minimum) for deep sedation and every 10 minutes (minimum) for moderate sedation.

4. after completion of procedure.

- For deep sedation, except pediatric patients < 30 kg. It is recognized that ETCO2 measurement may not be indicated during bronchoscopy.
- If maximum drug dosage is exceeded, the reason should be documented by the physician.

H. POST PROCEDURE ASSESSMENT

Postprocedure documentation must include: heart rate
blood pressure
respiratory rate
oxygen saturation
pain score
level of consciousness

Postprocedure observation must occur in a suitable location. Monitoring and documentation will continue every 10 minutes for 1 hour after the last dose of sedation or
until patient meets discharge criteria and/or returns to pre-sedation status. Patients receiving reversal agents will be monitored for a minimum of 2 hours. If discharged, the patient or responsible person must be provided verbal and written instructions regarding diet, medications, activities, and signs or symptoms of complications with course of action to take if any complication develops.

i. Post procedure / Post sedation Recovery Guidelines- Adult

Patient Sign	Criterion
Consciousness	Awake, responds easily, Responds readily, but easily falls asleep
Respiratory	Breathes easily with adequate volume. Slightly decreased rate and/or volume.
Circulatory	BP and pulses within baseline limits. BP and pulses approaching baseline limits
Activity	Able to move extremities voluntarily or on command (or returned to baseline)

ii. Post procedure / Post sedation Recovery Guidelines- Pediatric

Patient Sign	Criteria
Consciousness	Awake or returned to baseline Responding to stimuli
Respiratory	Breathes easily with adequate volume
Circulatory	BP pulses within baseline limit or approaching baseline limit.
Activity	Moving limbs purposefully (or returned to baseline)

I. <u>Sedation of Patients in ICCU</u>:

i. All patients admitted in ICCU will be sedated in such a manner that will

a. enable tolerance of endotracheal Intubation.
b. enable tolerance of mechanical ventilation.
c. enable pain relief in such a way that they are calm and yet arousable.
d. ensure that the patient is not aware of any procedures done.

ii. All patient on endotracheal intubation and mechanical ventilation will be managed as follows :

a) Inj. Midazolam 2 mg slow IV as initial bolus followed by 1 mg IV every hour. The dosage will be reduced over a period of time. During procedures a bolus

of 2 mg will be administered.

b) Inj. Diazepam 4mg slow IV followed by continuous infusion @ 1.2 mg/hr will be done if the patient is restless, struggling with the ventilator or markedly hyperventilating.

c) If the patient continues to resist the ventilator then Inj Thiopentone 0.5 gm IV infusion will be given.

d) Muscle Relaxant- Inj. Vecuronium 6mg slow IV bolus (0.1 mg/kg), followed by 1mg every hr will be the last resort.

Rescue of patient from deeper level of sedation:

iii. Rescue of a patient from a deeper level of sedation than intended is an intervention by a practitioner proficient in airway management and advanced life support.

iv. The qualified practitioner corrects adverse physiologic consequences of the deeper-than-intended level of sedation (such as hypoventilation, hypoxia and hypotension) and returns the patient to the originally intended level of sedation